

Icicle: Scalable Metadata Indexing and Real-Time Monitoring for HPC File Systems

Haochen Pan, Ryan Chard, Song Young Oh, Maxime Gonthier, Valérie Hayot-Sasson,
Geoffrey Lentner, Joe Bottiglieri, Rachana Ananthakrishnan, Kyle Chard, and Ian Foster

University of Chicago, Chicago, IL, USA

École de technologie supérieure, Montréal, QC, Canada

Argonne National Laboratory, Lemont, IL, USA

Purdue University, West Lafayette, IN, USA

Motivation

HPC file systems have outgrown the tools that monitor them.

Sites store 10s of billions of files across 100s of PB — find and du are $O(N)$ walks that take days at this scale.

Indexers like GUFi and Brindexer query faster, but remain batch-oriented: periodic scans miss the live state of the system.

Different consumers need different freshness: anomaly detection wants seconds, usage trends tolerate days; one rate doesn't fit all.

ORNL Lustre	Files per system	Single traversal
700 PB	10^9+	48 hr
Single file system	Tens of billions of objects	Reported on 10 PB Spider, 2010

Approach

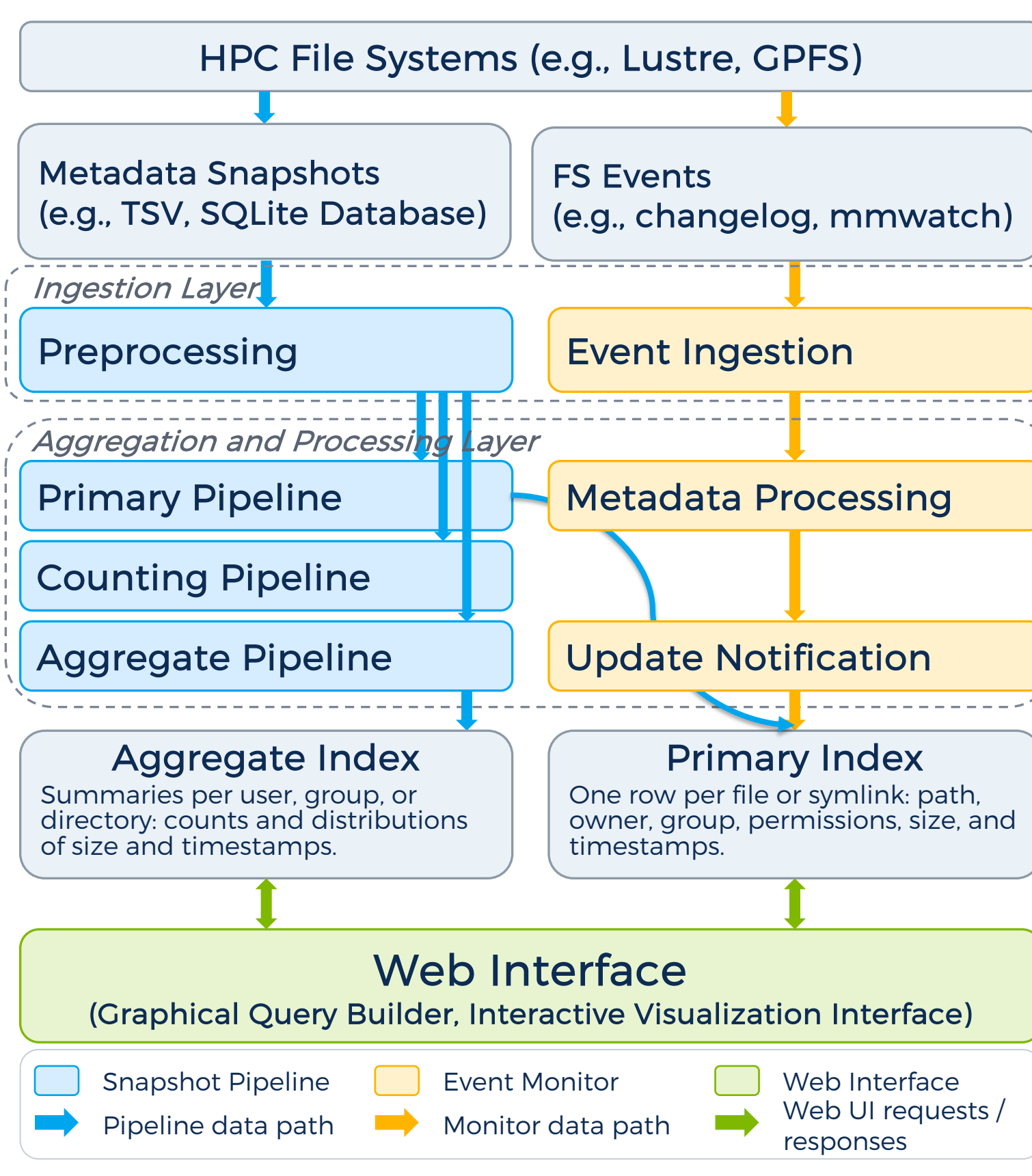
Icicle: a unified framework for continuous metadata indexing and monitoring.

1. **Unified schema.** One data model reconciles snapshots, GUFi databases, and Lustre / GPFS event streams.
2. **Streaming pipeline.** Kafka + Flink — idempotent, fault-tolerant aggregation.
3. **Production-scale evaluation.** Multi-PB snapshots; 4–100× over FSMonitor on Lustre.

Example queries Icicle answers:

Which large files have low access frequency?	Total storage consumption per project?	99th-percentile directory size per project?
--	--	---

Architecture



Datasets & Preprocessing

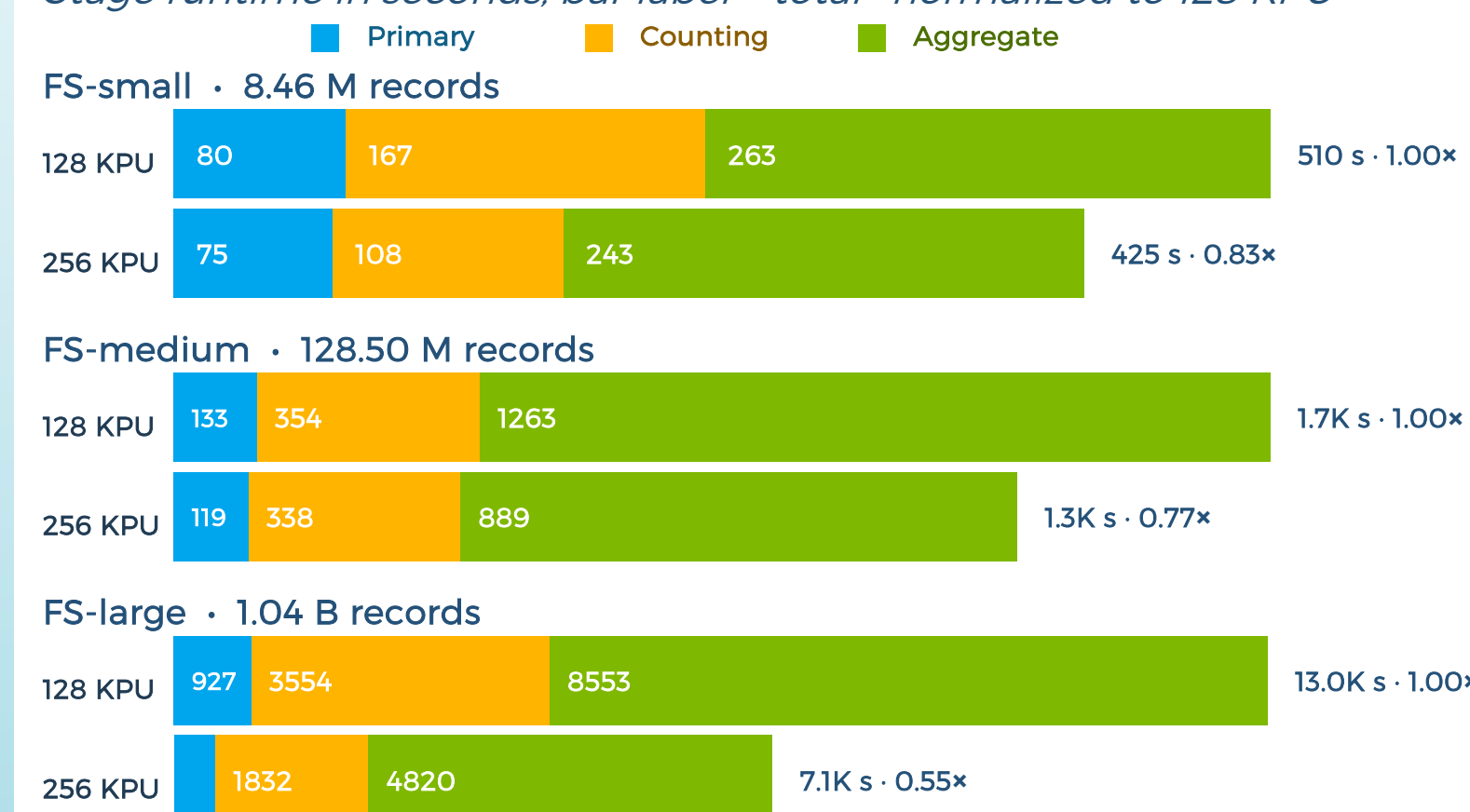
Three real production file systems — preprocessing cuts metadata up to 98.5%.

FS-small	FS-medium	FS-large
FILE SYSTEM SIZE 67.63 TB	FILE SYSTEM SIZE 1.55 PB	FILE SYSTEM SIZE 53.59 PB
METADATA SIZE 90.75 GB → 1.34 GB -98.5%	METADATA SIZE 38.83 GB → 21.68 GB -44%	METADATA SIZE 2.15 TB → 194.63 GB -91%
PRIMARY INDEX Records: 8.46 M Size: 5.78 GB	PRIMARY INDEX Records: 128.50 M Size: 90.85 GB	PRIMARY INDEX Records: 1.04 B Size: 774.83 GB
AGGREGATE INDEX Users: 37 Groups: 12 Dirs: 1,133 Size: 3.48 MB <i>1 level into user homes</i>	AGGREGATE INDEX Users: 240 Groups: 178 Dirs: 65,190 Size: 193.17 MB <i>2 levels into user homes</i>	AGGREGATE INDEX Users: 2,091 Groups: 325 Dirs: 16,724 Size: 56.95 MB <i>1 level into user homes</i>

Pipeline Runtime

Pipeline Runtime by Stage

Stage runtime in seconds; bar label = total · normalized to 128 KPU



8 min (FS-small) to 217 min (FS-large) on 128 KPUs;
7 min (FS-small) to 119 min (FS-large) on 256 KPUs.

- **Scaling:** 256 KPUs reduce FS-large runtime by 45%; small inputs are S3-bound.
- **Granularity:** re-chunking FS-small from 1 M-row CSVs to 100 K-row CSVs cuts total runtime by 46%.

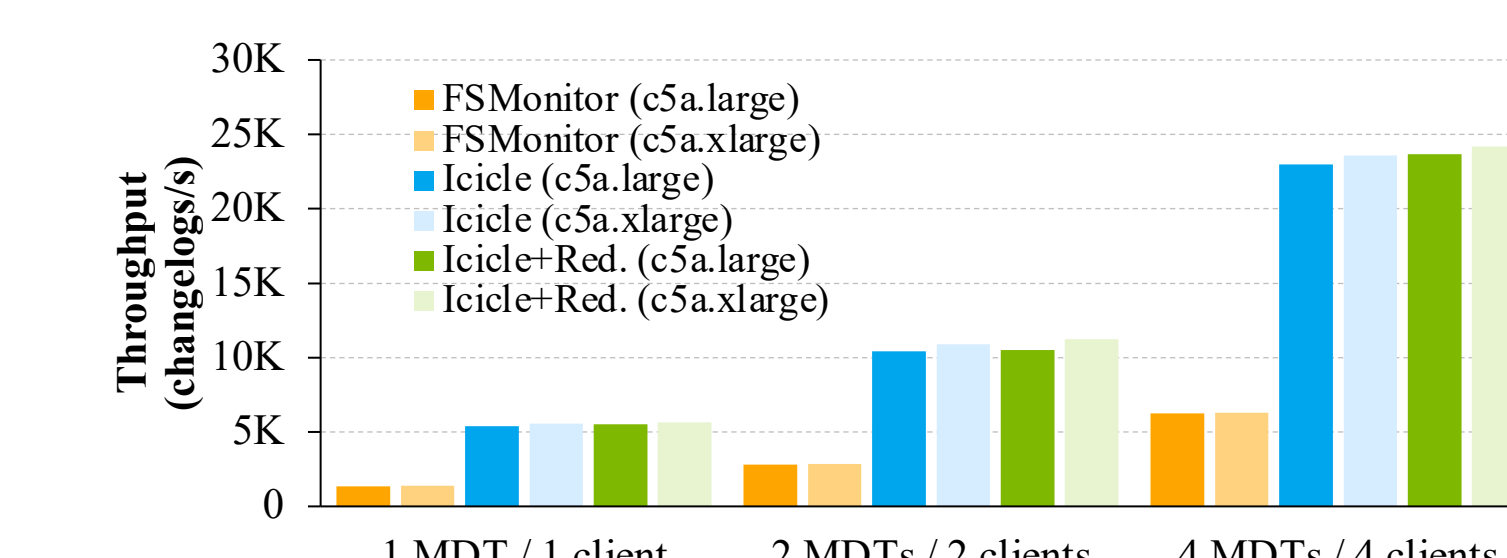
Monitor — Lustre

Near-linear MDT scaling.

- **Throughput.** 32–41 K changelogs/s vs. 391–565 for FSMonitor — 57–83× speedup.
- **Why.** One fid2path per directory; descendants resolved from cached parent state.
- **Scale-out.** 5.4 K → 23 K changelogs/s across 1 → 4 MDTs (Filebench).

Filebench Throughput vs. Lustre Cluster Size

Icicle+Red. = Icicle with event reduction enabled



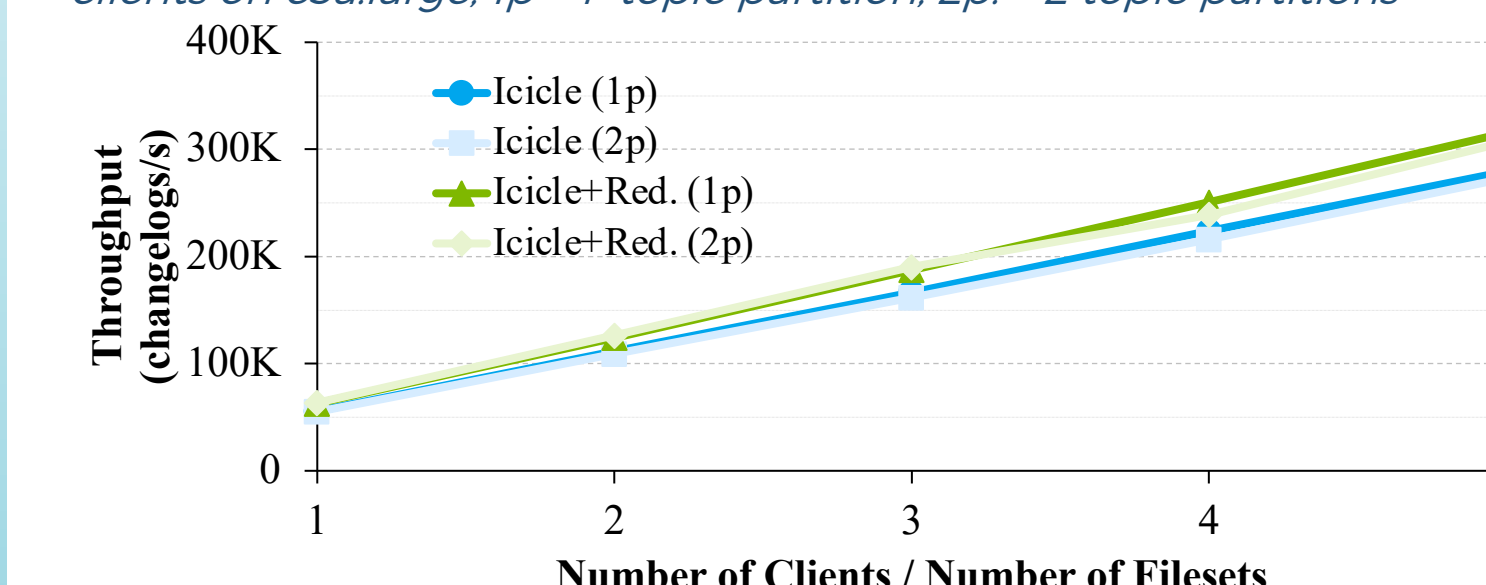
Monitor — GPFS

Linear scaling across filesets.

- **Throughput.** 55.7 K (1 fileset) → 279.5 K changelogs/s (5 filesets).
- **Why faster.** mmwatch embeds stat in events — no per-file stat calls.
- **Bottleneck.** Client CPU + state manager, not GPFS server.

Filebench Throughput vs. GPFS Cluster Size

clients on c5a.large; 1p = 1 topic partition, 2p = 2 topic partitions



Takeaways

Preprocessing is the scalability lever. Trimming metadata up to 98.5% cuts pipeline runtime to hours, and chunking aligns Managed Flink with S3 read parallelism.

Approximate where it's safe. DDSketch holds value error below 0.01 at ~2× the throughput of exact quantiles.

The event format and cache dictate throughput. On Lustre, deferring fid2path and reusing directory state delivers 4–100× over FSMonitor; on GPFS, mmwatch ships stat inline — eliminating a syscall per event and giving GPFS a further edge.

Paper



arxiv.org/abs/2604.10295

Code



github.com/globus-labs/icicle/